

CNAN

$$N = 32$$

1. Total DRAM element Access
$$= 2N^3 = 2(32)^3 = 65536$$

DRAM Traffic

$$= 65536 \times 4 = 262144 \text{ bytes} \approx 256 \text{ kB}$$

2. Tiled

$$\text{Per tile step} = 2T^2$$

$$\text{No. of tile steps} = \left(\frac{N}{T}\right)^3$$

$$\text{Total loads} = 2N^3 / T \approx 8192 \text{ loads.}$$

$$\text{Bytes} \approx 8192 \times 4 \text{ bytes} = 32 \text{ kB.}$$

3. Ratio

$$\text{Ratio} = \frac{256 \text{ kB}}{32 \text{ kB}} = 8 \neq N$$

The ratio would be equal to N only if it's an ideal tiled model, meaning the entire matrix dimension of 32×32 is reused,

4. Compute Time

$$\text{Total Flop} = 2N^3$$

$$\text{For } N=32 \Rightarrow 2 \cdot 32^3 \text{ Flops} \approx 65536 \text{ Flops}$$

$$t_{\text{compute}} = \frac{65536}{10 \times 10^{12}} \approx 6.55 \text{ ns}$$

Naive Case

$$\text{Naive DRAM Traffic } 256 \text{ kB}$$

$$\text{DRAM Bandwidth} = 320 \text{ GB/s}$$

$$\Rightarrow t_{\text{naive}} = \frac{256 \text{ kB}}{320 \text{ GB/s}} \approx 0.819 \text{ ns}$$

$$\text{Compute Time} = 6.55 \text{ ns}$$

$$\text{Memory Time} = 0.819 \text{ ns}$$

$\Rightarrow$ Memory Bound.

Tiled Case.

Case $\rightarrow 8 \times 8$ Loop count.

$$t_{\text{tiled}} = \frac{32 \text{ kB}}{320 \text{ GB/s}} \approx 0.102 \text{ ns}$$

$\Rightarrow$ Memory Bound.

Both Cases are memory Bound.